



UCL

BARC0141: Built Environment Dissertation

“A comparative study between isovist simulation and observation of pedestrian movement in the indoor context of an academic institution”

by

Sajid I Awal

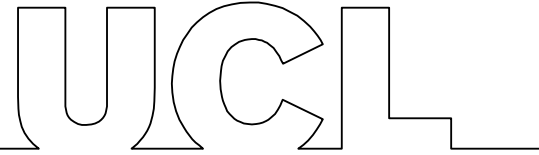
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Abstract

Although navigation in an indoor context is a widespread research interest, the research gap lies in combining real-life observation and computational simulation as methodologies. This research aims to contribute to such a research gap along with essential topics like indoor emergency evacuation that considers both the pedestrians with and without the knowledge of the context. To achieve such a result, 'movement trace' of space syntax theories, as the qualitative and isovist simulation as quantitative methods, has been used in this research. The findings evidence that the presence of destination points or goals impacts the difference between these two methodologies. Furthermore, the size of the context, the lines of sight or distance viewed, the number of hidden lines from the position of the pedestrians, the number of pedestrians and the number of obstacles to be avoided also influence such differences in the comparison.

Keywords

Indoor navigation, isovist simulation, pedestrian movement, observation, space syntax

Word Count

8484

Introduction

Many researchers have been interested in the navigation process of pedestrian movement. These studies are widely popular in an urban context. On such a bigger scale, navigation often saves pedestrians time while moving around the context to find the shortest paths to reach the desired destination from the starting location. This wayfinding or navigation process is also based on the field of view of the pedestrian and the obstacles they tend to avoid along their way. Nevertheless, the importance of navigation also prevails in smaller indoor spaces. Indoor navigation is crucial in predicting and assessing the pathways for emergencies like fire evacuation context. In such emergencies, the pedestrian, with and without the knowledge of the context, plays an essential role in enhancing evacuation efficiency (Tan et al., 2014; Marzouk et al., 2022). Although it is often hard to distinguish these two user groups indoors, this research aims to pick a context where they prevail. An academic institution has been taken under consideration for experimenting with this study, which satisfies this variety of user group conditions by having temporary exhibition space and space for regular use, where the exhibition space would include both pedestrians with and without the knowledge of the context, the space for regular use would only incorporate the users who have the thorough understanding of the context.

However, previous studies of such indoor navigation have been limited to qualitative or quantitative methodologies, particularly in dynamic contexts like academic institutions with two distinctive space types. Here, qualitative methods often followed several methodologies from space syntax theories to observe the movement of pedestrians in the indoor context (Wiener et al., 2007; Grajewski et al., 2001). On the other hand, several algorithms have already been explored in terms of navigation in indoor space, specifically with emergency evacuation in mind (Niu et al., 2016; Was et al., 2010; Natapov, 2022). However, the research gap lies in comparing real-life scenarios with the simulated results. Hence, due to the lack of a combination of these distinctive methodologies, generating a comparison between them, as shown in the works of Koutsolampros et al. (2017), is the intention behind this research under the theme of indoor navigation. Several study domains, like emergency evacuation, wayfinding, and pedestrian movement, have been considered for this research as direct relevance to this domain prevails in the research gap. Henceforth, to contribute to this domain of studies, this research aims to explore the navigation process in a selected academic indoor context to understand the role of simulation using an algorithm and evaluate the result based on the real-life observation of the context.

For this research, it is essential to observe the movement of pedestrians in the selected context, with a starting location that can also be used for quantitative methods and, eventually, the comparison. Hence, the methods for observing pedestrian movement in a selected context area, known as 'movement trace', have been used for the qualitative aspect. Whereas the algorithm for viewing the context from a particular pedestrian position, known as isovist, has been used for the quantitative method. Additionally, multiple pedestrians have been used to predict the complexity of each comparison. Factors like destination points and obstacles to avoid along the path have been considered in the quantitative simulation. Besides, two different space types based on the activities, temporary exhibition, and regular use of the space on the same floor plan of the selected context have been considered for this study. Finally, the qualitative and quantitative research results have been compared based on a selected grid size to see the similarities and dissimilarities between these two different space types. To keep the simulation realistic, wandering agents as pedestrians in the exhibition context have been used from a selected pre-defined start point, whereas for regular use, a destination and starting location have been pre-defined. The pre-defined areas are defined in both cases based on the observation result. So, the qualitative method has been implemented at the beginning of the research, based on which the quantitative approach has been explored for this comparison study.

Following these methodologies, intending to contribute to the domain of emergency evacuation of indoor context, navigation and pedestrian movement, the research objectives are to find the factors that eventually play a role in generating the difference between real-life observation and simulation. This research also hypothesizes that factors like complexity and openness of the context influence the similarities between these two methods. Additionally, the fixed destination location in some instances, like the regular use of the context, would have more common paths than the random wandering pedestrians in the context while comparing these. To address these two hypotheses, the research question for this research is -

'What factors influence the difference between isovist simulation and real-life observation of pedestrian movement in an indoor academic context?'

Background of Research

To address the research question centring the factors that influence the differences between isovist simulation and real-life observation of pedestrian movement in the selected indoor academic context, the background of this research has been divided into two major parts. First, extensive studies have been conducted on pedestrian movement by looking into emergency evacuation, navigation, and agent-based simulation, specifically in the indoor context. Then, studies on relevant methodologies regarding qualitative analysis using space syntax methods and isovist simulation regarding quantitative methods have been reviewed. The methodologies would clarify the algorithms' definitions and the reasons behind the parameters.

1. Literature Review

Fewer studies have been conducted concerning building layouts and pedestrian route choices (Tong et al., 2022). In addition to that, due to the complexity of indoor environments, the demand for navigating in such a context is also increasing (Yan et al., 2021). Wayfinding, pathfinding, or navigation is the process of accurately finding the pedestrian location and reaching the goal followed by a route (Jayakanth et al., 2020). Wayfinding is widely popular in the urban context, and researchers have also practised indoor navigation despite needing more practice due to its complexity (Yan et al., 2021). The author also pointed out that most navigation systems in the indoor context offer a single source for the shortest distance or fastest paths, which is ineffective in contexts like shopping malls, museums, or hospitals (ibid.). Several factors can influence navigating due to the dynamics of pedestrian behaviour, like gender, age, and physical ability, and three challenges in modelling such context are understanding pedestrians, data, validation, and verification (Marzouk et al., 2022). Although wayfinding is a popular research topic to be explored with various methodologies, Khotbehsara et al. (2023) only used qualitative space syntax methods in the context of hospitals with the impact of the pandemic. Grajewski et al. (2001) discussed the methodology from space syntax theories on observing and tracking pedestrian movement called 'movement traces', which illustrates the path pedestrians take in indoor spaces. Besides, Montello et al. (2023) focused on the difference between symmetric and asymmetric route choices regarding the pedestrian movement of wayfinding. As decision-making is essential in wayfinding, space syntax methods are often used qualitatively to optimise the results, as studies conducted by El-Darwish (2022) show. Regardless, some research also explored the domain using virtual reality technologies to consider other senses and visuals (Lyu et al., 2023).

In indoor navigation studies, knowledge of space the pedestrian plays a vital role in forming their desired path (Tan et al., 2014). Pedestrians with and without knowledge of the context can influence the indoor wayfinding process, and this diversity is related to the efficiency of emergency evacuation (ibid.). In an indoor context, most research that uses computational simulation as a quantitative

method for wayfinding surrounds the context of emergency evacuation of the building (Niu et al., 2016; Was et al., 2010; Natapov, 2022). In the evacuation context, pedestrians without specific behaviour would follow several factors, like the distance that has already been travelled in the room to select the exit location (Liu et al., 2011). Variation in the knowledge of the context, the interaction between pedestrian and static objects also varies, making the emergency evacuation process more challenging (Marzouk et al., 2022). Besides, a self-organizing map (SOM), a supervised machine learning algorithm commonly known as the travelling salesman problem, has been used by Yan et al. (2021) in an indoor context where the shortest path does not satisfy the demand of the context's uses, as pedestrians tend to visit selected locations in the space. The agent-based simulation is a popular preference while modelling pedestrian movement, particularly in the case of emergencies in an enclosed indoor environment. In contrast, Kasereka et al. (2018) and Luh et al. (2012) used Computational Fluid Dynamics to model the fire and an agent-based model as pedestrian evacuation. Said et al. (2012) considered the geometrical components of the building along with the shortest path in such simulation. Alternative to agent-based simulation, Burstedde et al. (2001) used cellular automata to model pedestrian simulation along with social force models in the evacuation context., Pasandi et al. (2021) used MASA or a modified A* algorithm for similar simulations. Along with agent-based simulation, obstacle avoidance is essential to emergency evacuation and wayfinding research domain. Ni et al. (2020) considered a multi-obstacle environment for a multi-navigation evacuation simulation. Cui et al. (2023) used Newton's law of motion and an obstacle avoidance algorithm to simulate a similar context. Considering both agents and static objects as an obstacle is also considered an essential part of this research domain, as shown in the works by Song et al. (2006). Formolo et al. (2017) could not consider obstacles like columns or multiple rooms while considering only the social and psychological factors of the agent's behaviour. However, isovist simulation, based on the longest visible distance from the pedestrian's position, is also used for navigation simulations that represent real-life navigation based on visibility (Koutsolampros et al. 2017; Wiener et al. 2007).

2. Background on Methodology

Both qualitative and quantitative methods are used in this research. The qualitative method includes observing the pedestrian movement from the space syntax theories known as 'movement traces'. The quantitative methods of simulating pedestrian movement in the context are based on isovist simulation algorithms. Finally, an evaluation criterion is used to compare the results derived from qualitative and quantitative methods to determine the factors influencing the difference between these two methodologies. The background on these two selected methodologies and their evaluation process is described below briefly –

i. Qualitative Methodology Background – Movement Trace

Space syntax theories are widely popular in observing pedestrian movement in both outdoor and indoor contexts, where indoor context observation provides a more controlled environment with a selected group of pedestrians based on the context. Space syntax was initially meant to analyse large-scale spatial configurations from indoor to outdoor space in terms of the scale of the context (Wiener et al., 2007). The term 'space syntax' refers to the set of tools consisting of simple graphs with paths and nodes that analyse the spatial configuration of the context (ibid.). Fan et al. (2022) used three indicators of space syntax theory, namely accessibility, visibility, and intelligibility, to demonstrate the need for physical access, visual access, and spatial cognition in pedestrian movement. One of the findings noted that the urban context has negligible visibility (ibid.). This visibility factor has been considered for this research as a quantitative method; isovist simulation is based on this. Besides, Chun et al. (2019) previously used 'static snapshot' methods from the space syntax methodologies to study users' choice of seats in a particular indoor context. The author considered the types of activities conducted by the users during observation and collected data for

over two consecutive days for over 14 hours for this study, where some of the data collected resulted from human error (ibid.).

However, this research is based on the 'movement trace' from space syntax theories, as Grajewski et al. (2001) mentioned. Here, the aim is to generate the path the pedestrian used during their presence in the context to compare it with the quantitative method results later. This methodology is expected to have 75% accuracy. The 'movement trace' method includes standing on a particular point for a fixed amount of time, usually 3-5 minutes, marking the precise route taken and locating the person last seen. Here, only a particular space within a more prominent space is selected for the study to understand better the chosen area inside the more prominent space (ibid.). Additionally, this method will give the starting point of the pedestrian, the average distance the pedestrian travels, and the movement pattern from the starting point. Here, although it is possible to approximate the number of pedestrians who visited the space being observed in the selected period, it is unlikely to determine whether the pedestrian knows the area, specifically when they are in the context where both types are authorized to have access.

ii. Quantitative Methodology Background – Isovist Simulation

Many different approaches and algorithms have been experimented with for wayfinding in indoor contexts. Several models like the social force model (Helbing et al., 2000), floor fluid model (Burstedde et al., 2001), and multi-grid model (Song et al., 2006) to simulate the pedestrian as homogeneous particles moving around as emergent functions of several forces in respect to the context (Tan et al., 2004). This research focuses on the isovist simulation used by Koutsolampros et al. (2017) and Wiener et al. (2007) for similar research in an indoor context. A brief description of this isovist simulation as the quantitative method, along with its definitions and parameters used in this research, are as described below in brief –

Isovist simulation was introduced by Benedikt (1979) to analyse spatial characteristics of the indoor context. It uses viewshed polygons from the observation point of the pedestrian to measure the visible area of the context (Wiener et al., 2007). Turner et al. (2011) further developed the methodology by introducing the intervisibility of multiple observation points in the context of the efficiency of the simulation. Koutsolampros et al. (2017) used isovist simulation and generated a grid for evaluation in similar wayfinding research. Factors like the walking speed, cone of vision, and obstacles to avoid are essential to consider in this simulation as the algorithm's parameters. In the study conducted by Ha et al. (2012), physical contact between a pedestrian and another or any static object is considered an obstacle. Similar conditions can be applied to static obstacles like walls or furniture. Besides, this obstacle avoidance is activated inside the cone of vision, which is, ideally, 170 degrees for each pedestrian (Gjestland, 2020; Tan et al., 2014).

Additionally, the starting position of the algorithm and the destination location can also be used as essential parameters, where the destination location is often considered as the attractor point, with an attraction force towards which the algorithm will allow the pedestrian to move by avoiding the obstacles on the paths. These starting and destination locations are the same as found in the qualitative method to keep the common factor between the comparison.

iii. Evaluation Criteria Background

Evaluating the results achieved from both the qualitative and quantitative methods is an essential part of this research. It would determine how far the quantitative methods are off from the real-life scenarios to achieve further improvement. More precisely, it will point out the factors that influenced such differences, which was the main intention behind this research. Previously, Koutsolampros et al. (2017) used a grid system to find the activated cells while comparing the various simulation types. In

the study by Burstedde et al. (2001), the building environment is geometrically represented as grids consisting of small cells of 40 by 40 cm or 400 mm by 400 mm, which is approximately the typical space occupied by a pedestrian in a dense crowd. The authors claimed it influenced the simulation result and defined the pedestrians' maximum density and flow rates for such simulations (Lord et al., 2005; Pelechano et al., 2008). The cells overlapping with obstacles like walls or inaccessible spaces are considered occupied in the research conducted by Tan et al. (2014). Finally, the results are compared quantitatively by counting the number of activated cells in these two types of methodologies.

Research Methodology

The research methodology is divided into four parts to identify the factors influencing the difference between isovist simulation and real-life observation of pedestrian movement in an indoor academic context. Firstly, the research context is outlined, consisting of two different types of space that serve the complexity of having both pedestrians with and without the knowledge of the context. Secondly, the qualitative method is clarified, consisting of 'movement trace' methods used for the selected context. Thirdly, the isovist simulation method is elaborated along with the pseudo-code used. Lastly, the evaluation criteria method used to compare the qualitative and quantitative methods has been clarified. A brief elaboration on these respective parts of the methodology is as follows -

1. Research Context

To bring variation in the data sets of this research and support the arguments stated in the research background, both the pedestrian with and without the knowledge of the context have been considered. This makes the selection of research context a vital part of this research and is essential as it is also considered in the indoor evacuation context research domain as the efficiency often partially depends on the knowledge level of the pedestrians (Tan et al., 2014). The selected context for this research is the 2nd floor of the Bartlett School of Architecture building of University College London, UCL, located at 22 Gordon Street in London, UK. Here, two different space types have been considered for two separate studies. In comparison, one space consists of public access and is often used as a temporary exhibition space and needs more complexity. The other space is a part of the studio space used regularly throughout the year by the users with the knowledge of the context. In the exhibition space, pedestrians with and without knowledge of the context are allowed access. Still, that space is temporary and isn't the same after the exhibition. Hence, these two space types have been selected to explore different factors of quantitative methodologies against the same qualitative method that requires a limited space within the observer's view from a location.

The study on the temporary exhibition space was conducted during the Bartlett Summer Show 2023. During this exhibition, both visitors, as the pedestrians without the knowledge, and the regular students and staff of the institution, as the pedestrians with the knowledge of the context, were allowed access. The exhibition was carried out across all six building floors on the same zone on every floor, keeping the studio space inaccessible to visitors. The main staircase was used as the primary mode of vertical circulation and central entrance point to this selected study space. Although, in some cases, disabled persons were noticed to be using elevators to access this context. The following figure shows the chosen floor plan's temporary exhibition context and location.

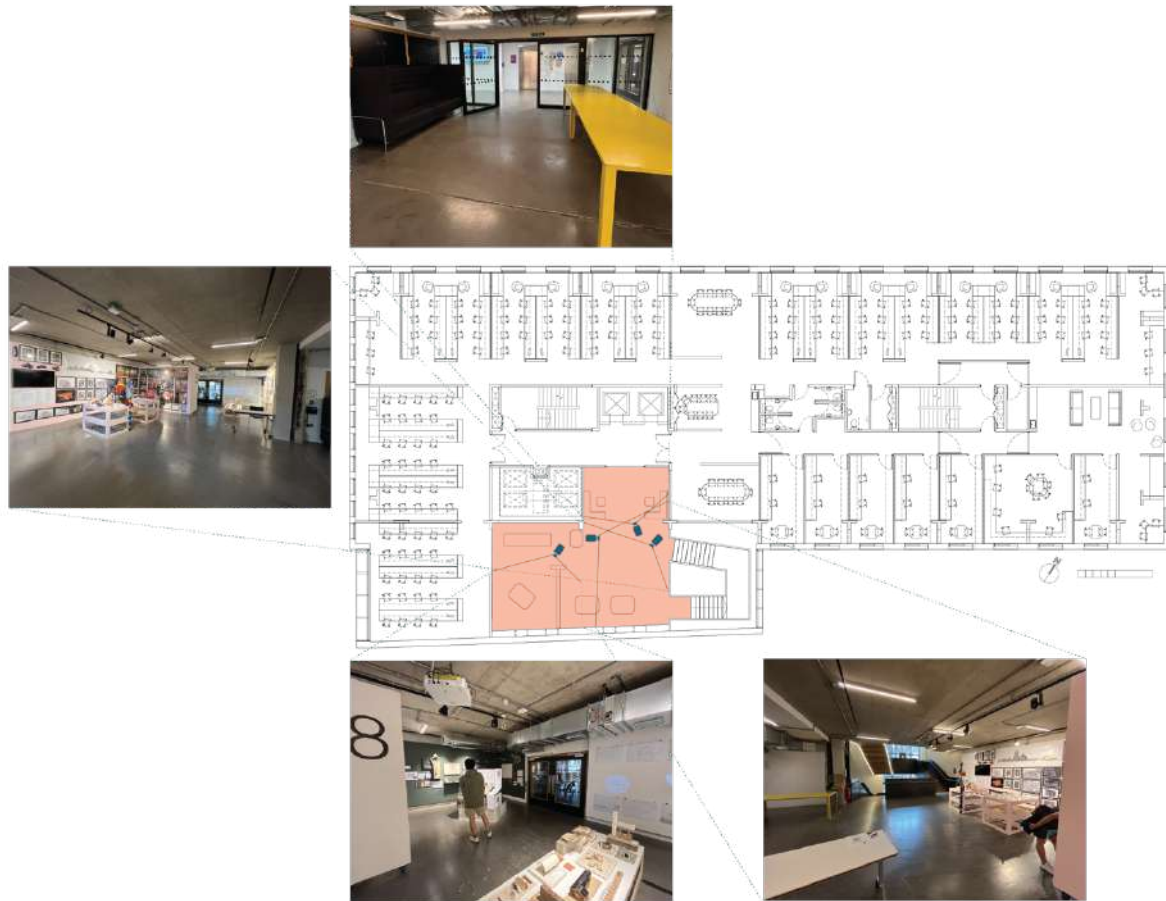


Figure 1: Indoor temporary exhibition space as context at Bartlett Summer Show 2023, UCL (2nd Floor)

The regular studio space is another type of space that is most used throughout the building. This space is typical and identical across all the floors other than the ground floor. A part of the space is selected after entering the door from the elevator, often restricting access to only the students and staff. Hence, this context is suitable for the group of pedestrians who know the context. Here, studios are almost open, with partial partitions, without doors, and situated alongside the main corridors that serve as the central circulation. The following figure shows the chosen floor plan's studio space selected for the study on the regular use of the context.



Figure 2: Regular use of the studio space as context at Bartlett, UCL (2nd Floor)

2. Qualitative Methodology

Movement of pedestrians in both the temporary exhibition and studio space for regular use has been observed using the same 'movement trace' method from space syntax theories clarified in the research background. To keep consistency in the qualitative methodology of these two space types with different uses, only this method has been selected, and thus, there is limited space for observation than the whole floor plan. The observation in the studio space with regular uses has noted the starting and destination points as the pedestrian often has a destination location as the goal. In comparison, only the starting location has been observed for the temporary exhibition space. Here, the pedestrians' behaviour is more like wandering in the context of enjoying the exhibition without any specific goal or destination location.

For the temporary exhibition, the research was conducted for 15 hours over three consecutive days with three time slots over 5 hours each. The period included from 10 am to 3 pm on days one and two, and the last day had from 3 pm till 8 pm until the exhibition ends to cover the whole time. On day one, the overall setup of the context, potential busy times, and the number of pedestrians accessing the space were observed. The movement of pedestrians was marked on a typical floor plan, highlighting the exhibition space in focus. Both the pedestrians with and without the knowledge of the context were observed. However, it was not possible to distinguish them. However, the time frame was not considered, as different pedestrians stayed and moved in this space for further periods as they intended to view the exhibits. On the other hand, the studio space with regular use has been observed for over two consecutive days for 20 hours. The period includes 9 am to 2 pm, then 3 pm to 10 pm daily. The part of the studio not visible from the observation position was considered out of the research context. In other words, the entrance location of the pedestrians for each studio was

regarded as the destination location or end point of the path. In both cases, a path was drawn on the floor plan separately for each group of pedestrians passing through the selected context.

However, due to the methodology's limitations, the pedestrians' generated path would be partially correct. Regardless, following this method, this research uses qualitative analysis to identify the approximate number of people visited in both contexts, their starting and end points, particularly for studio space, and preferred paths taken in the contexts. This information is later used to evaluate the research by comparing the generated path with the results from the quantitative method.

3. Quantitative Methodology

To keep consistency with the qualitative method results, the quantitative method using the isovist simulation has been divided into two types. For the temporary exhibition context, a wandering agent isovist simulation is used as the agent is predicted to wander in the space without any specific goal or destination location, but only a starting point. On the other hand, for the studio space with regular uses, along with the starting point, a destination point acting as an attractor has been used for simulating the agent's path. However, in both cases, obstacle avoidance has been kept consistent. The defined parameters used for this isovist algorithm, along with the generalized pseudo-code, are described briefly below -

Koutsolampros et al. (2017) and Wiener et al. (2007) used isovist simulation for pathfinding studies in the past; the method used in this research has been adapted for this study. For this research, a single pedestrian, five and ten of them, are considered for isovist simulation, and the parameters are defined following the literature review. The simulation avoids obstacles, including static objects like furniture, walls, and other pedestrians, as Ha et al. (2012) noted. The pre-defined starting point has been used for this isovist simulation in both contexts, and the simulation has been run for several minutes. The algorithm has been run using C# programming language inside Grasshopper. The depth parameter that generates the number of intersections with the context has been limited to 30 due to the room size. The cone of vision has been limited to 170, which is the ideal cone of vision for humans, as noted by Gjestland (2020). For obstacle avoidance, the Rhino. Geometry. Intersect. Intersection. CurveCurve method has been used, whereas Curve. CreateInterpolatedCurve method has been used for visualizing the path travelled by the pedestrian in a particular time frame from the starting point to the endpoint. The algorithm has used Flock and IsoAgent, where the IsoAgent class contains all the calculations, including isovist and obstacle avoidance. In contrast, the flock class uses the IsoAgent to generate the desired simulation for a flexible number of pedestrians.

The pseudocode for the isovist simulation is described below in a generalized format. Despite being written in C#, the following pseudocode can be adapted to any programming language.

1. Declare and initialize the necessary variables and data structures.
2. Check if birds are null or if the reset is true:
 - If true, execute the following steps:
 - Initialize the rnd variable with a new Random object.
 - Create a new flock object (birds) with the specified parameters.
 - The flock object should have a list (birdsList) to store the agents.
 - The flock object should initialize the agents with the specified parameters.
 - Continue to the next steps.
3. Update the flock:
 - Call the UpdateFlock method of the bird's object with the specified parameters.
4. Initialize lists for storing data:
 - Create an empty list (boidsLocationList) to store agent positions.
 - Create an empty list (boidsTrailsList) to store agent trails.

- Create an empty list (isovistRaysVisuals) to store agent data.
- 5. Iterate over the agents in the flock:
 - For each agent in birds. birdsList, execute the following steps:
 - Add the agent's position to boidsLocationList.
 - Add the agent's data to isovistRaysVisuals.
 - If the agent's trail has more than 2 points:
 - Add the agent's trail curve to boidsTrailsList.
- 6. Assign values to output variables:
 - Set isovistRays as isovistRaysVisuals.
 - Set updatedAgentLocation as boidsLocationList.
 - Set agentTrail as boidsTrailsList.
- 7. Define the Flock class:
 - Define properties and methods to manage a flock of agents.
 - Implement methods to update the flock and perform flocking behaviours.
- 8. Define the UpdateFlock method in the Flock class:
 - Accept parameters for controlling the flock's behaviour.
 - Iterate over the agents in the flock and update their positions and behaviour accordingly.
- 9. Define the Agent class:
 - Define properties and methods to represent an individual agent in the flock.
 - Implement methods to observe the environment, update the agent's state, and calculate isovists.
 - Create an instance of the Flock class with the desired parameters.
- 10. Iterate for the desired number of iterations:
 - Call the UpdateFlock method of the flock instance.
 - Update any necessary variables or data structures.
- 11. Use the outputs (isovistRays, updatedAgentLocation, agentTrail) for further processing or visualization.
- 12. Define additional helper functions or classes as needed.
- 13. Call the Main function or start the program execution.

4. Evaluation Criteria

The evaluation of qualitative and quantitative methods consists of comparing the number of grid cells activated by them and their common grid cells. The reasons behind their difference are the main aim of this research. Hence, for evaluation, a grid consisting of 400 mm by 400 mm has been generated to represent the selected context geometrically for evaluating qualitative and quantitative methods, followed by the research conducted by Koutsolampros et al. (2017). Since Burstedde et al. (2001) used a 400 by 400 mm grid, the same grid has been followed for the visualization for this research. The number of cells occupied or traversed by pedestrians has been considered activated cells. On the contrary, the cells consist of obstacles like walls and left-out furniture, as Tan et al. (2014) noted.

Research Experiment

The experiment for this research has been divided into two main parts based on two separate selected contexts consisting of different uses and types of pedestrians and their comparison. One is the regular use of studio space accessible to pedestrians who know the context, mainly the students and teachers of the institution. Here, the pedestrians usually have a destination location or a goal, which they tend to reach from their starting position. On the other hand, the temporary exhibition space, to which pedestrians with and without knowledge of the context, students and visitors are allowed to have

access and visit this space during the event period. Here, pedestrians have more freedom and tend to move accordingly without any fixed destination location.

Henceforth, for this research, pedestrian movement is observed and simulated with a fixed destination point for the context of regular use. In contrast, more free-flowing pedestrian movement is observed and simulated for the exhibition context. Then, the results are compared for the evaluation to derive the factors influencing the difference between them. Besides, in both cases, simulation is done with single, five and ten agents, respectively, who have the same starting point and goals, depending on the context, but still avoid each other and static obstacles along their path while performing their role. One experiment is elaborated and compared for each type of experiment, regular use, and exhibition context. Additionally, the other seven conducted for each type are described collectively, along with significant findings.

1. Regular Use Context

The qualitative analysis was conducted first to generate the destination and the starting position of the pedestrians in the selected context for regular use, which is a part of the studio space. Around 50 and 30 people were observed on the first day, and around 40 and 10 people were observed on the following day in two different time slots mentioned in the methodology section. Then, following the destination point, the isovist simulation have been generated to try and match the path observed for each path taken by the pedestrian. Due to the simulation's flexibility, five and ten pedestrians are simulated to see the difference as if five or ten students were visiting the exact location or studio space with the same starting point. Additionally, the drawings of the context have been made simpler with less furniture and single lines as outlines representing obstacles for ease of simulation and computational load.

The following diagram shows the path generated by a single pedestrian derived from the qualitative method, highlighting the starting and endpoints. However, the direction has yet to be shown in this diagram and research. The result progresses from left to right with one, five and ten pedestrians, respectively, in the isovist simulation, with the same path generated from observation. All three of the simulations avoid obstacles and each other for multiple pedestrians but follow a similar path by remaining close to each other. The parameters have been kept the same for every three simulations. At the endpoint, all three simulations tend to move a bit further than the end goal, which is unrealistic because of the attraction force that has yet to be set for this research.

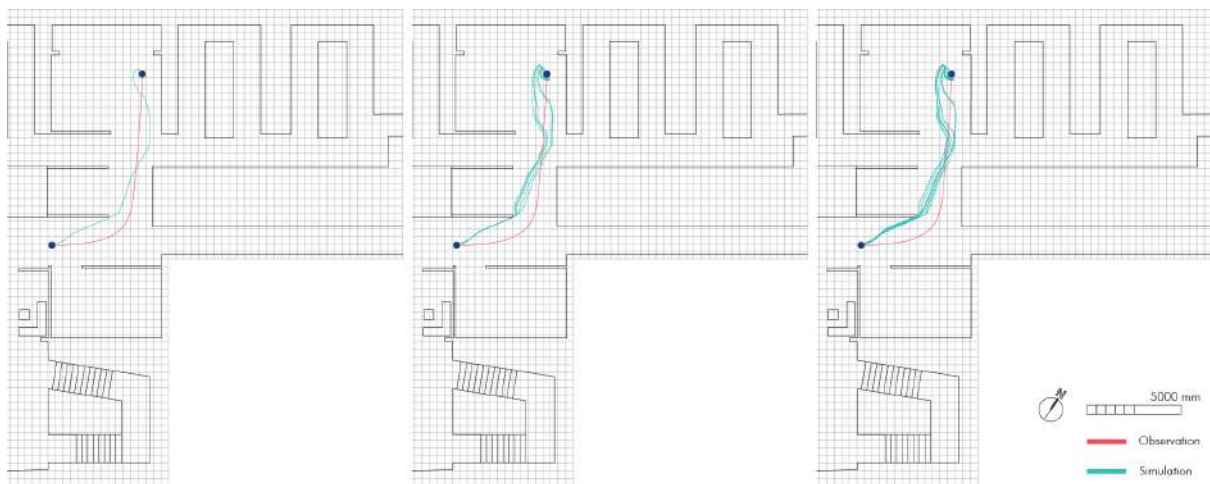
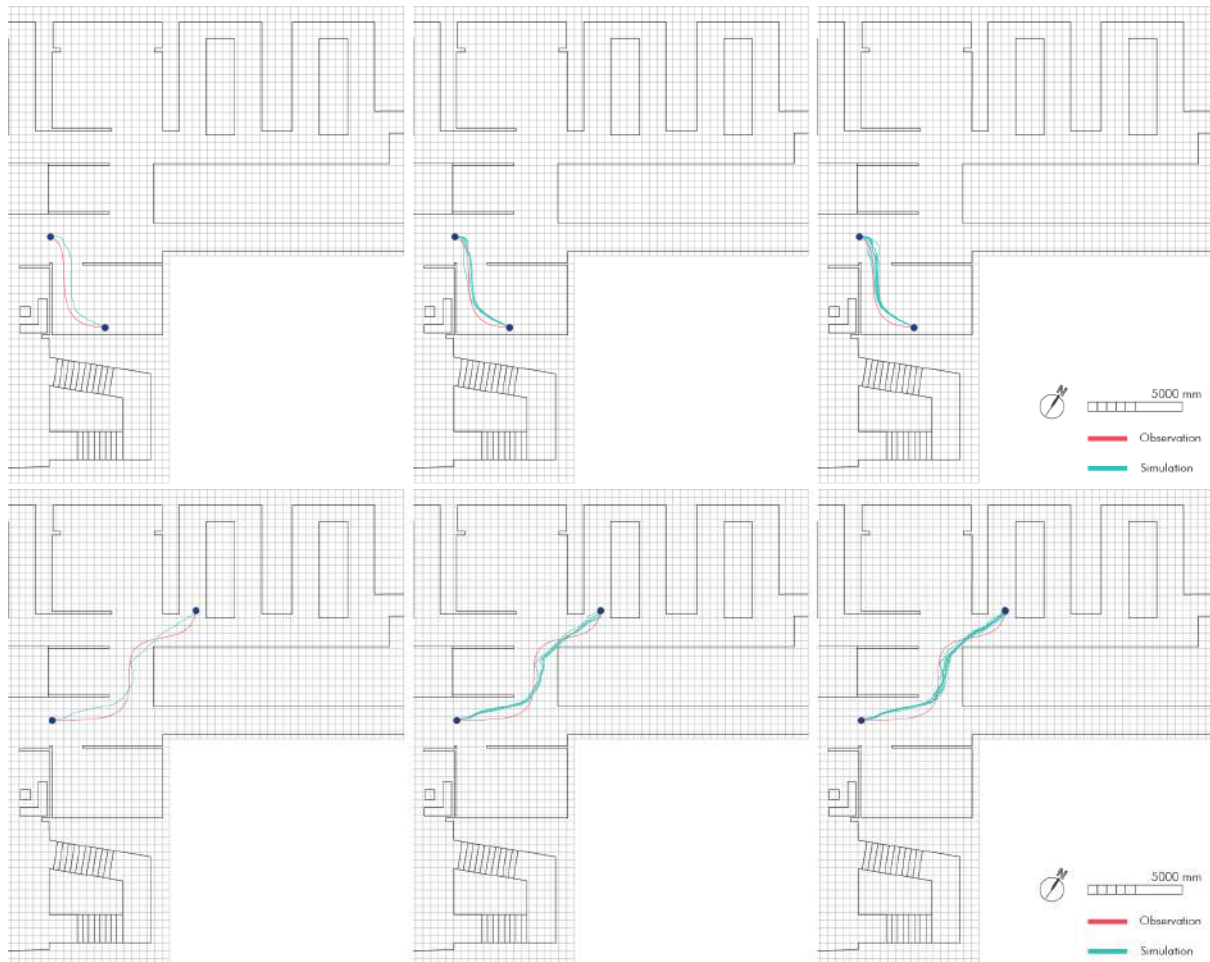


Figure 3: Observation and simulation of one of the pedestrian experiments in regular use of the studio space at Bartlett (2nd Floor)

Similarly, seven other experiments were conducted, each following the observation starting and destination. Here, the destination point is the entrance to each studio space in the selected context. The format is the same as the previous diagram, with one, five and ten agents simulating respectively on the left in proper chronological order. The furniture in the studio space has been removed for faster simulation results. Here, fewer obstacles must be overcome for the pedestrians, so the simulation result seems quite close to the observation result. The parameters have also remained the same from the previous diagram. Still, regardless, due to fewer obstacles being avoided, the simulation tends to come closer to the observation than the analysis on the previous diagram. The starting point is also the exact location of the previous analysis. The simulation and observation paths overlap in the central path, meaning they have taken a similar route. These can be clarified when comparing the routes for the evaluation. Here, the longer the distance, the more obstacles avoidance in the simulation are, the more accessible and the closer to the observation that can be proven when compared for evaluation. For longer distances, some of the pedestrian paths were not avoiding obstacles, which can also be considered as one of the drawbacks. The reason is that some obstacle lines were hidden from the starting location with an attraction force to the destination. Besides, one of the drawbacks is that the simulation had to be tested multiple times to generate the desirable result, meaning the path closer to the observation. The clustered or bundle effect is seen in multiple pedestrian simulations despite each avoiding the others as obstacles.





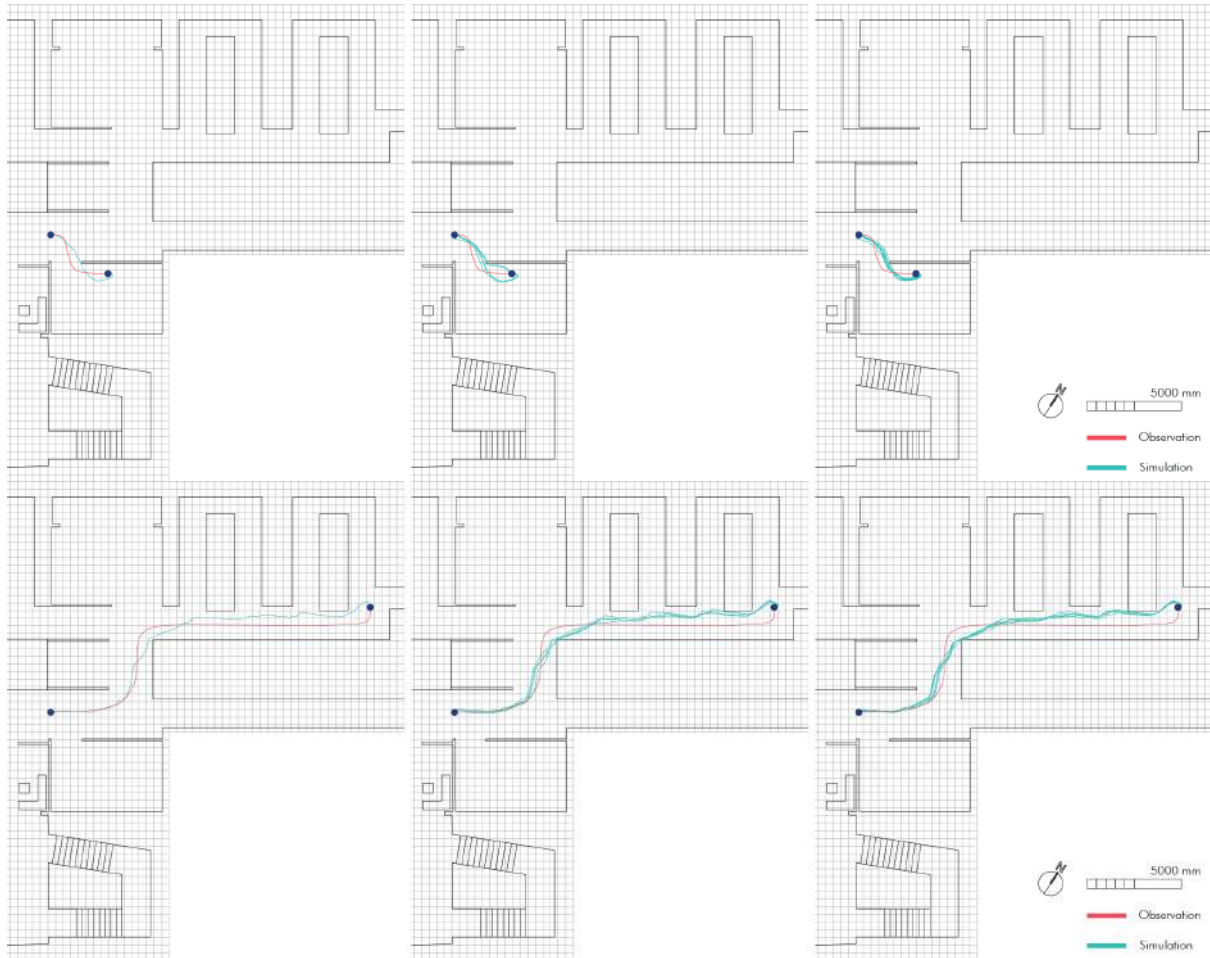


Figure 4: Observation and simulation of the rest of the pedestrian experiments in regular use of the studio space at Bartlett (2nd Floor)

2. Exhibition Context

A slightly different isovist simulation has been conducted for the temporary exhibition context. Here, the context is temporary, meaning the space is different after the exhibition period, but it brings both the pedestrians with and without the knowledge of the context. From the qualitative analysis using the same method as the regular studio space context, the number of pedestrians observed was around 20, 30, 40, 40, and 50 people on average per hour in the morning from 10 am to 3 pm. In contrast, the average number of visitors, including students and the public, from 3 p.m. to 8 p.m. was around 40, 60, 30, 20, and 10, respectively per hour. The observation also finds that the visitor aims to see the exhibit, unlike the students who already know the context. The students would prefer to check the specific project they preferred to view during their time in that space. However, it was only possible to distinguish the students and visitors if they were known or had been wearing identification and, therefore, not distinguished in this research. However, the context considered has these two types of pedestrian groups and is essential, as discussed in the literature review. Hence, the visitors would generally walk slower upon entering the room, covering the whole room, unlike the students. Besides, the exhibition path is pre-designed for easier circulation to cover the entire exhibit. So, the space that is visible at first glance upon entering the space would be reached first by the visitors, which is different for the students. Besides, the visitors would often travel in groups, unlike the students. Additionally, if one space is occupied, other pedestrians tend to avoid it. Hence, the obstacle includes other pedestrians prevailing in the space as well. Henceforth, more free-flowing wandering pedestrians have been considered for the isovist simulation that moves freely aimlessly in space by avoiding obstacles and following the longest distance hit from the point of observation.

Here, the eight observations and simulations are considered for this temporary exhibition context. Although the simulations have the same starting point as the observation, the nature of the simulation is left as wandering behaviour without any goal or destination locations. The following diagram shows both observations and simulations of the selected exhibition context. Here, all the observation paths were compared with each simulation type as the behaviour is not expected to have specific goal or destination points.

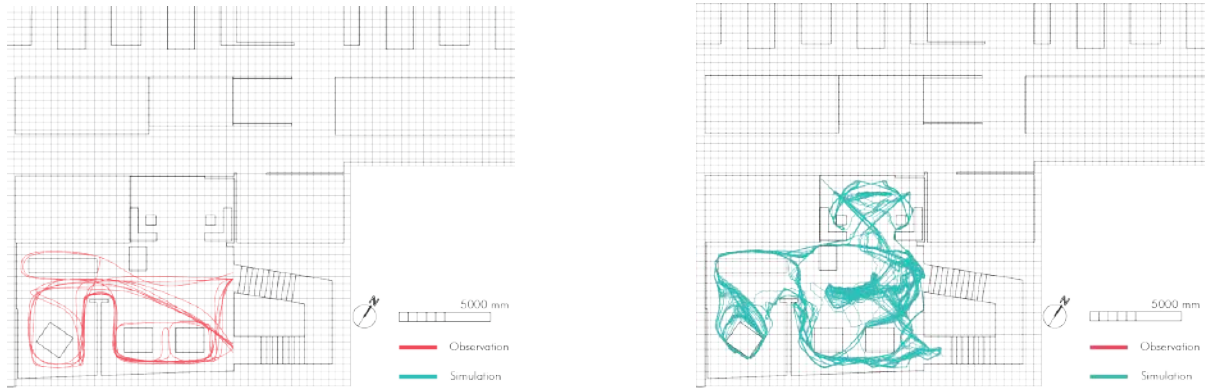


Figure 5: All the observation paths(left) and simulated paths(right) of the temporary exhibition context are shown here collectively.

From the observation, it is evident that the main staircase is considered the point of entrance to this space, and the pedestrian tends to move towards the exhibition and follow the circulation path that is kept along the exhibits and is unlikely to visit the other end that is leading towards the elevators and lacks any exhibits. On the contrary, for the simulation, the pedestrians tend to move towards the other side of the room where there aren't any exits, which is unrealistic, as evident from the observation. The reason is due to the lack of knowledge of the location of the exhibits in the simulation. However, numerous attraction points still need to be considered for this research. Besides, the central axis evident from the observation that connects the starting point to the other exhibition space is not visible from the main entrance point and hence lacks in the simulation results. Here, only a single starting point derived from the observation has been defined for the simulation.

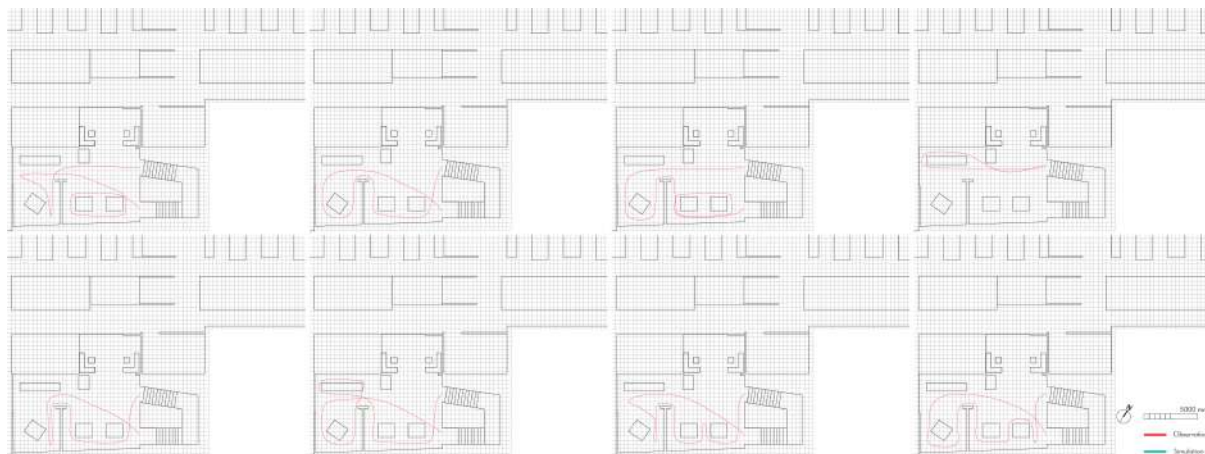
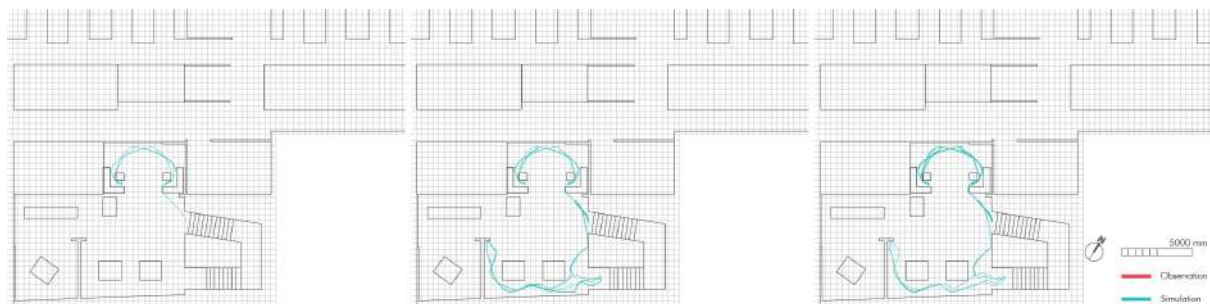


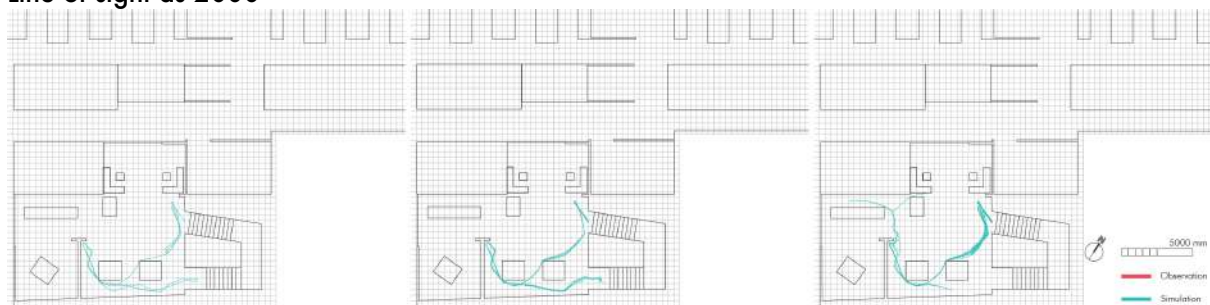
Figure 6: Eight separate observations for the temporary exhibition context

The following diagram shows the simulation with different lines of sight, which tends to affect the simulation behaviour; each is then compared with the collective observation of eight various pedestrian paths. Like the regular use context, from left to right, the number of pedestrians simulated is one, five, and ten, respectively. Multiple pedestrians in the simulation with the same starting points often show slightly different results than the single pedestrian, regardless of parameter changes. They

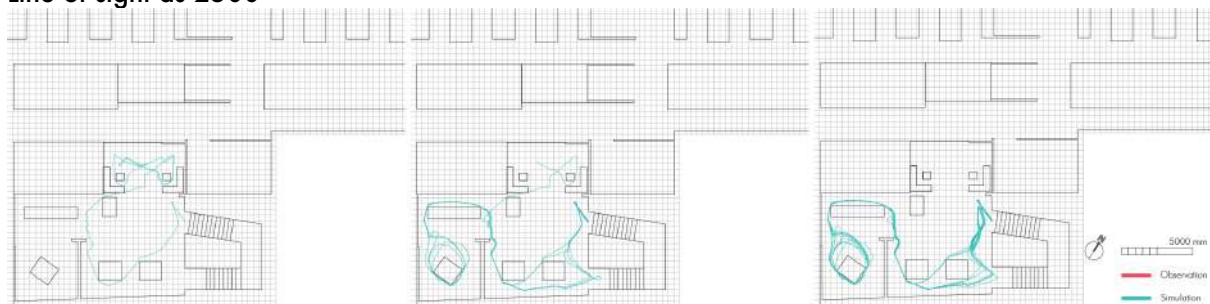
tend to divide and not be in a cluster or group, unlike the analyses conducted in the regular-use context. A few of the pedestrians in the simulation move to the other room, but the path taken is not like the observation. They tend to divide themselves from the origin point. A group tend to move towards the other side of the space towards the elevators by avoiding obstacles along the way and then visit the unvisited room by the single pedestrian simulation from the opposite direction. Here, the pedestrians in the simulation are moving following the smaller lines of projection that avoid the obstacles, as there are no attraction points or force in this simulation code. These lines of sight also act as the observation and are responsible for creating the wandering behaviour of the simulated pedestrians. Some pedestrians might follow the path formed by the observation, which cannot be determined as the time required for each observation path was not measured as it varies and hence cannot be used as a criterion for the comparison in this research.



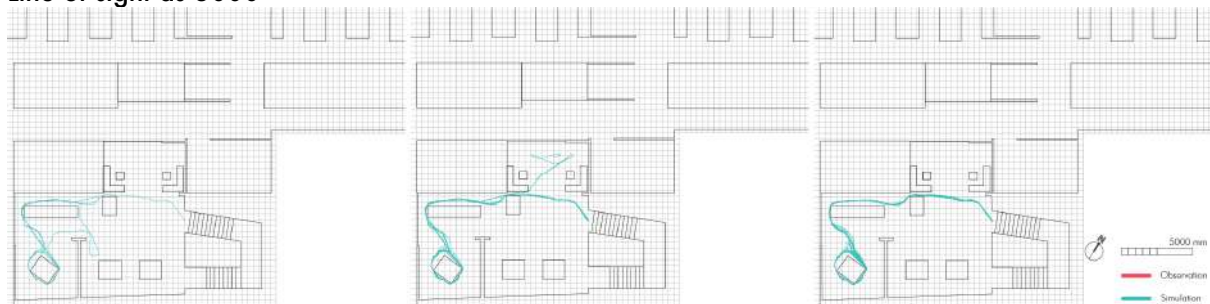
Line of sight as 2000



Line of sight as 2500



Line of sight as 3000



Line of sight as 3500

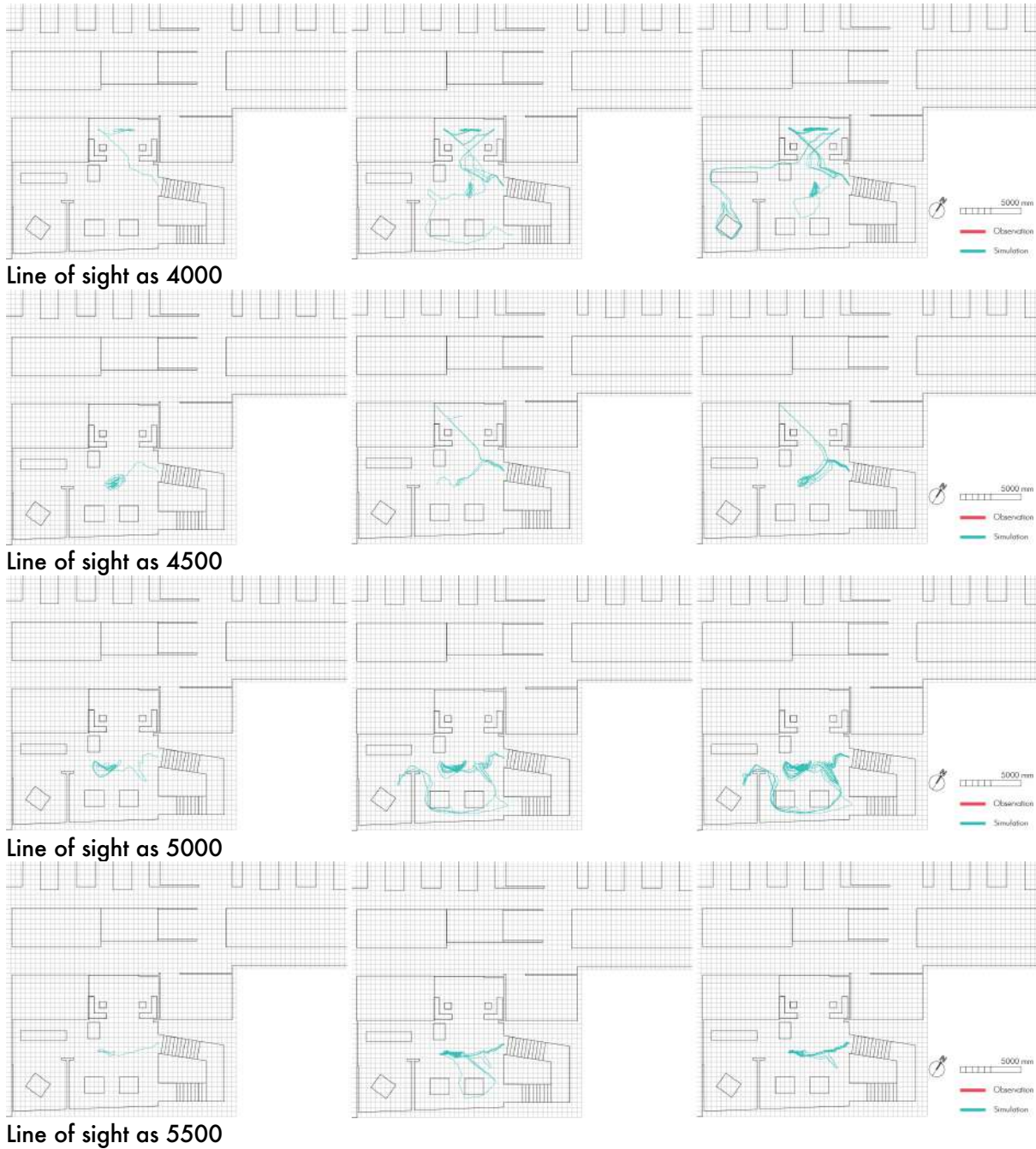


Figure 7: Eight separate simulations for the temporary exhibition context with different lines of sight

3. Evaluation

A grid is formulated as described in the methodology to evaluate the observation and simulation results for both space for regular use and the temporary exhibition context. Here, the activated cells in the selected grid would be the cells through which the paths have been passed. However, the regular use and temporary exhibition context comparison have been conducted separately despite having the same grid and process with activated cells. The reason lies in the different simulation methods used, where one has an attractor as the goal for the regular use context, and the wandering behaviour is tested out for the exhibition context. Hence, the number of activated cells would be counted between simulation and observation for the regular use space. On the other hand, all the

activated cells by all the observations would be compared against each simulation for the exhibition context due to variations in the line of sight in the parameters.

i. Regular Use Context Comparison

Following the results discussed in one of the pedestrian experiments of regular use, the following diagram shows the number of activated cells and the common cells between them for that particular experiment. In this regular-use context, it is evident that, for single pedestrian simulation, the result is closer to the observation, meaning that the number of activated cells in the quantitative methods is similar to that achieved from qualitative methods. However, the number of overlapping cells is minimal compared to the total. This implies that the path generated from observation using the qualitative approach differs from that of the quantitative method. Besides, when the simulation is multiplied by five each time, the number of activated cells increases along with the number of overlapping or common cells.

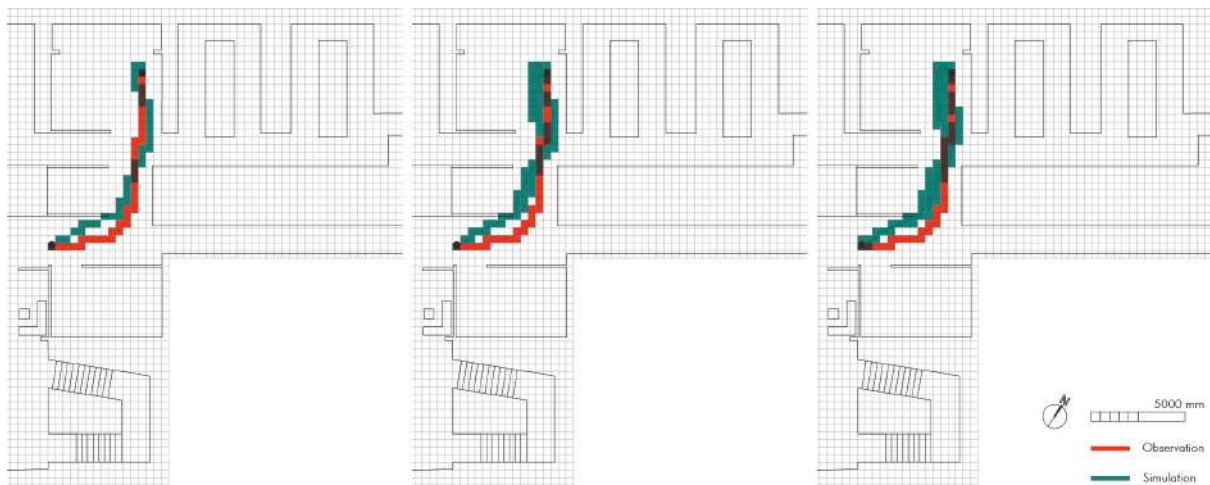


Figure 8: Comparison of observation and simulation of one of the pedestrian experiments in regular use of the studio space at Bartlett (2nd Floor)

With the information achieved from the comparison discussed above, the following diagram shows the percentage of common cells or overlapping cells between these two methods. It is noted that, compared with observation and single, five and ten pedestrian simulations, respectively, the number of activated common cells is 20.51%, 18.46%, and 21.92%. These statistics are almost identical in this case and are less than one-fourth of the total path taken. This implies that the path one or multiple simulations take when having a destination point or target is almost identical and does not vary with the number of simulated pedestrians. Here, it is also evident that the paths pedestrians take in quantitative and qualitative simulations are not the same, and the algorithm can only predict a certain percentage of the actual path taken by the pedestrian.

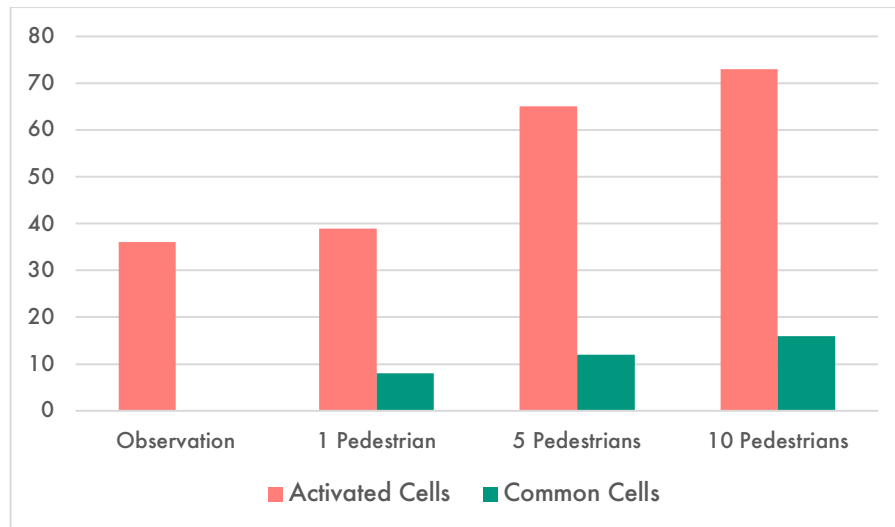


Figure 9: Comparison by number of activated and common cells between observation and simulation of one of the pedestrian experiments in regular use of the studio space at Bartlett (2nd Floor)

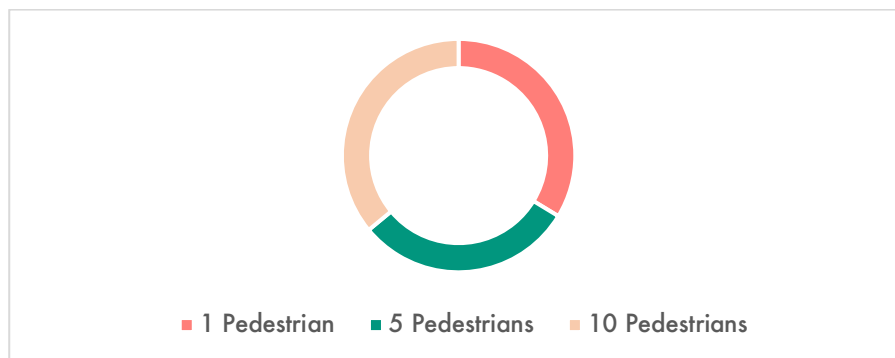
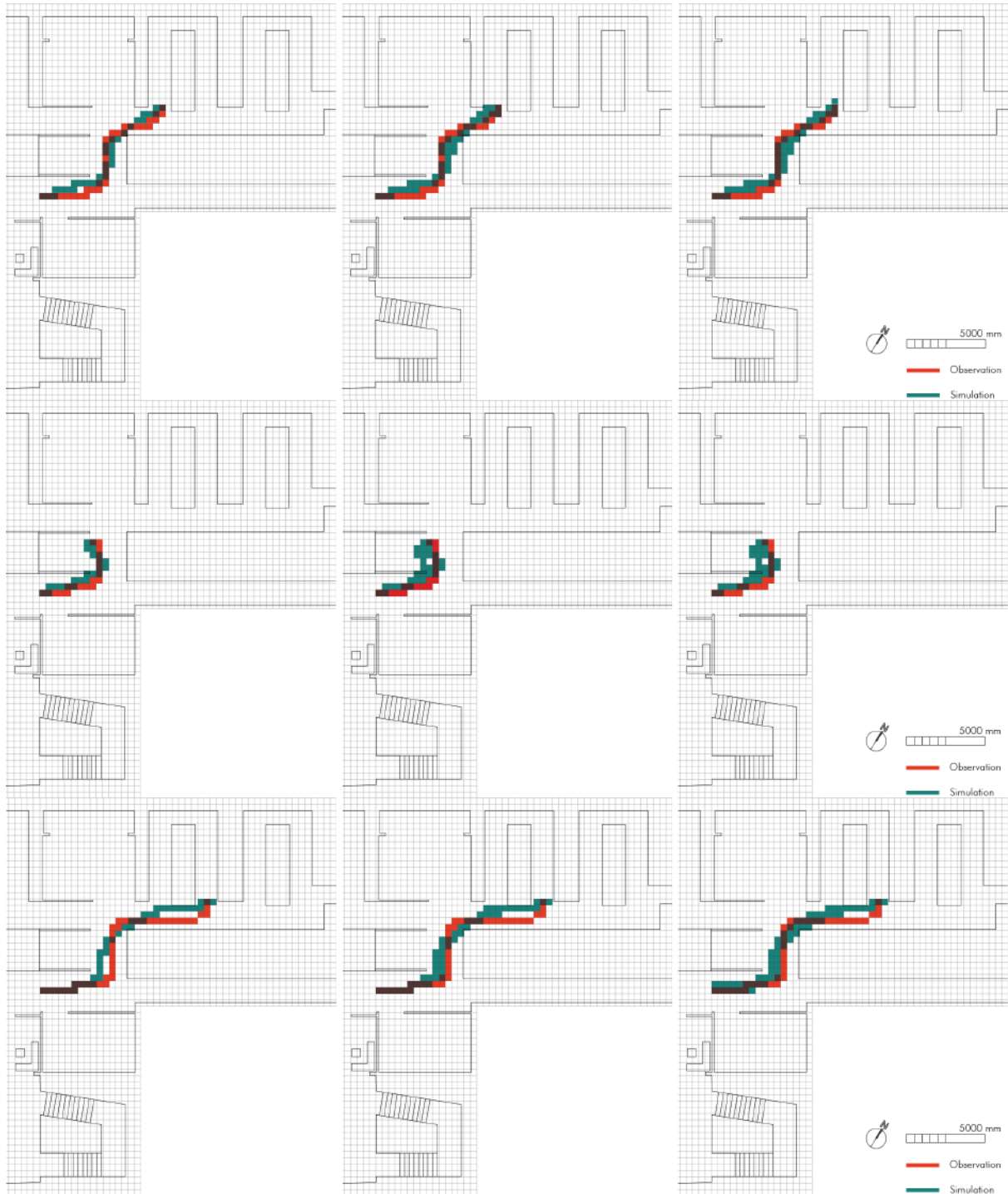


Figure 10: Comparison by the percentage of common cells between observation and simulation of one of the pedestrian experiments in regular use of the studio space at Bartlett (2nd Floor)

Similar methods have been used to experiment on the other seven comparisons between observation and simulation in the following diagram. Here, it is evident that the attractor plays a crucial role in bringing the simulation closer to the observation results. Context also plays an essential role in the differences between the activated common cells and the results of these two methodologies.





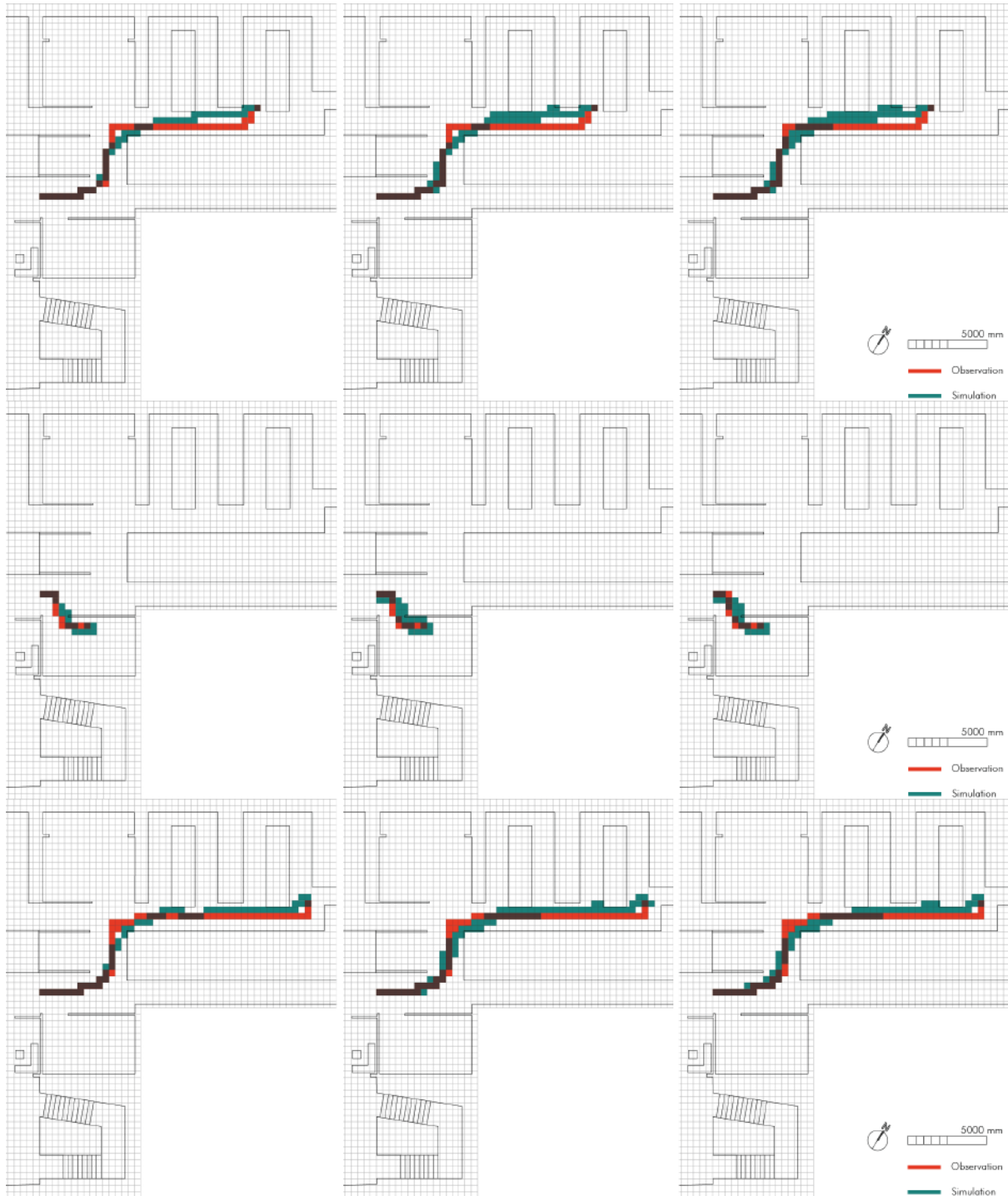


Figure 11: Comparison of observation and simulation of the rest seven of the pedestrian experiments in regular use of the studio space at Bartlett (2nd Floor)

ii. Exhibition Context Comparison

For the exhibition context, unlike the comparison done for the regular-use context, the number of activated and common cells are compared between all the observation paths and each of the simulations, where the simulation consists of single, five, and ten pedestrians, respectively, depending on the lines of sight used in the simulation as a parameter. Due to differences in comparison, unlike the comparison process conducted for the regular use context, counting the number of common cells would be insufficient to identify the desired findings. Since the simulation has no destination and goal,

the results differ from those in a regular-use context. Besides, the size of the context and the number of obstacles is also different, influencing the simulation results.

The following figures show the comparison between all the observations and all the simulations with various lines of sight conducted for the exhibition context. Here, the darker cells denote that those cells have been more activated or are common for each type of analysis. A significant difference is evident due to the nature of simulated code in this experiment, which is different from the regular use context, which had a destination point. For this exhibition context, the wandering pedestrians in the simulations moved to the other side of the context, which lacks the exhibits and is unrealistic. However, some common cells are also evident between observation and simulation when all of them are compared.

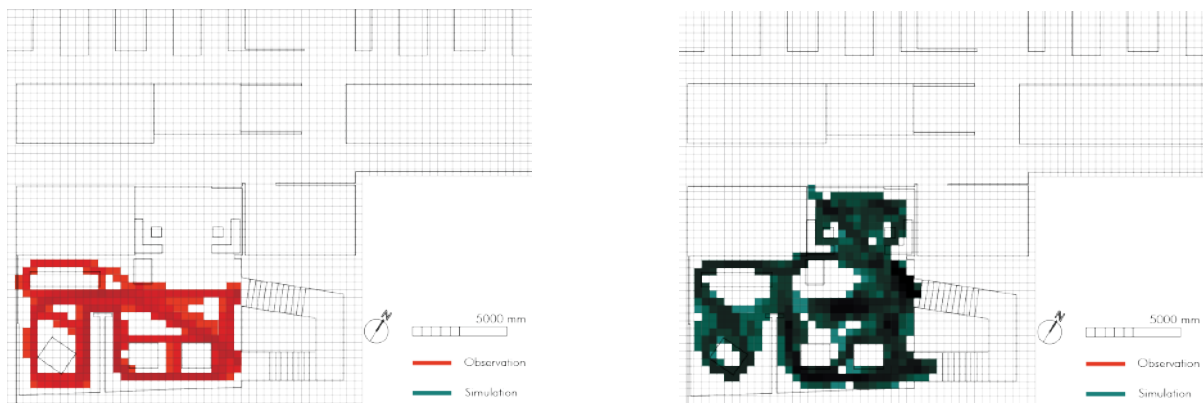


Figure 12: Activated cells in the observation(left) and simulations(right) in the exhibition context

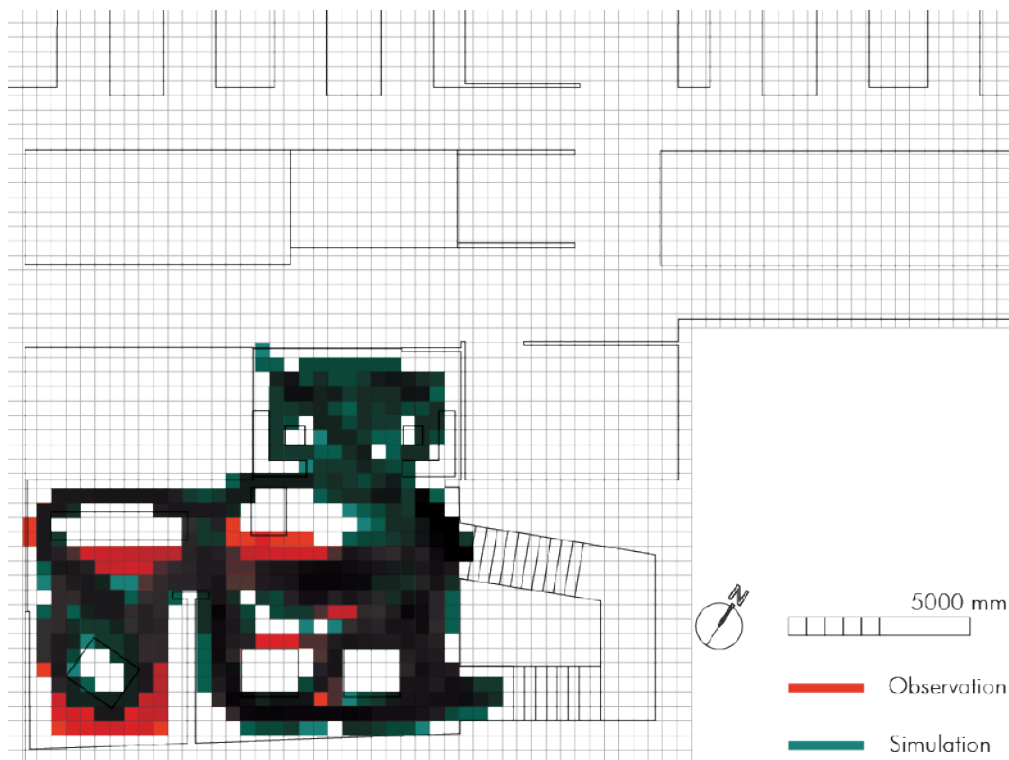
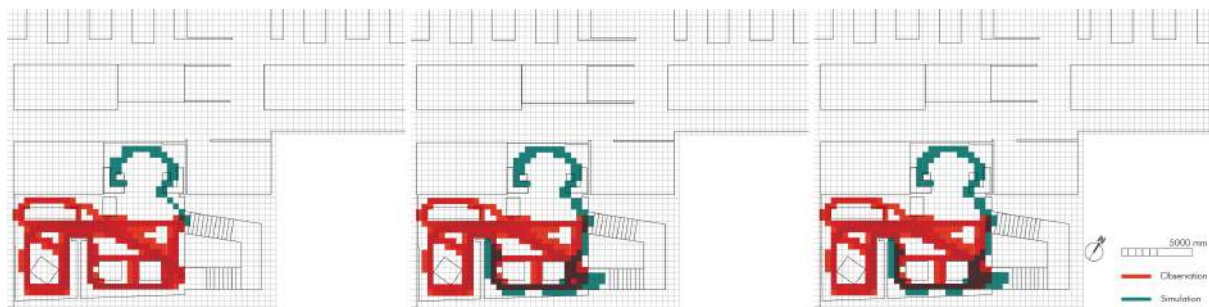


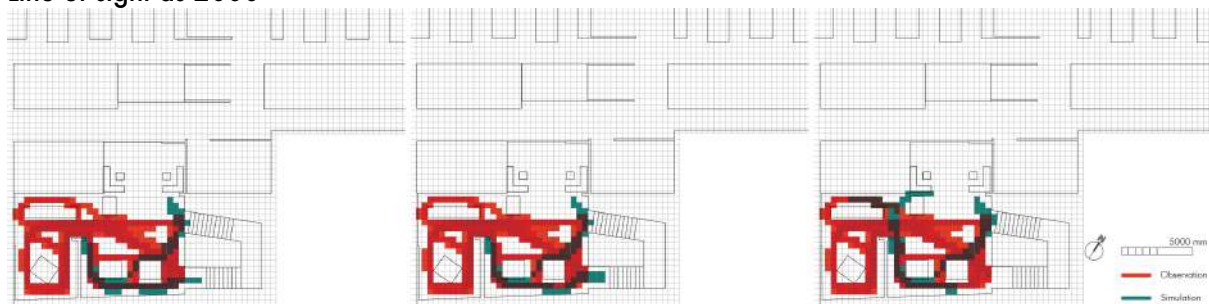
Figure 13: Activated cells in the observation and simulations in the exhibition context

The following diagram shows all eight simulations conducted with various lines of sight and compares each with all the observations. The result signifies the differences between these two separate

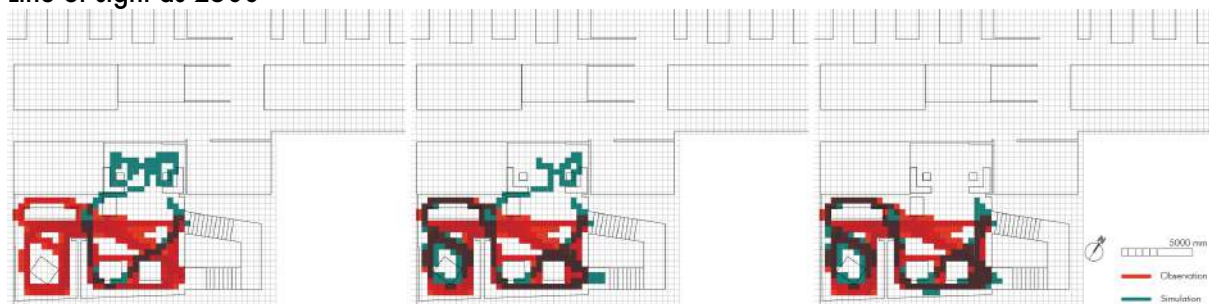
methodologies of qualitative and quantitative analyses. Unlike the regular use context, which is with a destination, here, when simulated without a goal, it is often seen that almost half of the route taken by a single pedestrian simulation is similar to the one found in observation. For multiple pedestrian simulations, the results are often identical to the observation paths. This implies that the path taken without having a destination point or target varies with the number of simulated pedestrians and comes closer to the observation when only a single pedestrian is simulated. Here, as seen in the previous result, it is also evident that the paths pedestrians take in quantitative and qualitative simulations are different, and the algorithm can only predict a certain percentage of the actual path the pedestrian takes. The comparison also shows that the number of activated cells increases when the number of simulated pedestrians increases. One of the major drawbacks from this experiment is the lack of destination point, that takes some of the simulation to the other side of the space, where no observation path was found due to absence of exhibits, and hence those simulations are unrealistic.



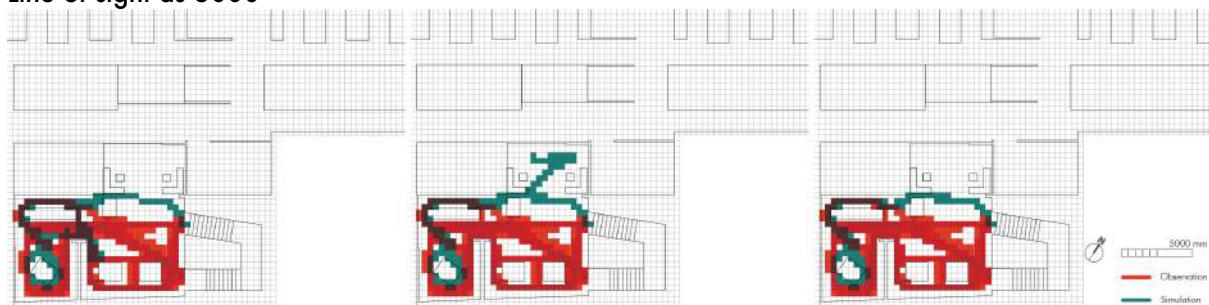
Line of sight as 2000



Line of sight as 2500



Line of sight as 3000



Line of sight as 3500



Figure 14: Activated cells in all the observations and eight different simulations in the exhibition context with various lines of sight

Research Discussions

Previously, Marzouk et al. (2022) noted that several factors influence navigating due to the dynamics of pedestrian behaviour, like gender, age, and physical ability, and three challenges in modelling such context are understanding pedestrians, data, validation, and verification. However, such factors were not considered for this research. Here, the research findings suggest that several other factors, apart from the dynamics of pedestrian behaviour, contribute to the difference in this context's selected quantitative and qualitative methodology. However, the quantitative method partially depends on the qualitative one in this research for finding the path's destination and starting point. Having a

destination and a starting point is a significant factor influencing the differences. The notable difference between pedestrian movement in the exhibition and regular use of indoor context prevails in the goal of the pedestrians. Whereas for exhibitions, the pedestrians aim to view the exhibits, and for regular use, they tend to take the shortest path to their respective working spaces. Besides, many points can be pre-designed for exhibitions as the goals represent attraction points that pedestrians tend to use. At the same time, regular use makes the pedestrian reach one specific destination, representing only one attractor as their workspace. Besides, the number of pedestrians in the quantitative method impacts the differences only when the destination point is not considered. The size of the context and the number of obstacles to be avoided also affect the differences. In the exhibition context, where wandering pedestrians have been simulated, lines of sight also play a crucial role in formulating the different types of results and the number of pedestrians being simulated. Partial clustering is evident for the exhibition context for multiple pedestrian simulations, whereas for regular use context, which has a particular destination as an attraction point, clustering for almost all the multiple pedestrians is evident. Distance also does impact the simulation, as some lines of the obstacles that are not visible from the starting location can affect negatively, meaning diverting the desired path for some pedestrians in multiple pedestrian simulations as seen in the regular use context. Regardless, the algorithm isovist for the simulation can only partially predict pedestrians' paths, as evidenced by the qualitative methods.

Despite its significant findings, several limitations in this research can be further considered. For instance, the direction of the moving pedestrians in neither qualitative nor quantitative analysis was considered, which might impact providing more information on the differences between these two distinctive methodologies. Besides, the two types of quantitative simulations, one with the destination position as an attraction point and one without such attraction point or force, were not considered in the same space that could also be considered for more information on the desired cause of the differences. Although the research finds obstacle avoidance in moving pedestrians and static objects, the interaction between pedestrians can be considered for further in-depth research. The time issue for the quantitative method used, where no attraction point or destination position was defined, can be addressed further. Here, multiple attraction points or destination positions could be placed so that the moving pedestrians move along these attraction nodes based on their weight values in the quantitative analysis. However, the time required to complete these paths in qualitative and quantitative methodologies still needs to be considered. Since people tend to spend different times in front of the exhibits, estimating the time spent in each node or taken for each path is impossible. In terms of qualitative methods, other methodologies like 'people following' or 'static snapshots' from space syntax theories can also be considered in addition to the 'movement trace' used in this research to generate a more refined qualitative analysis.

For future works, several other parameters, like the biological and psychological factors of the pedestrians and geometrical constraints of the context, can be considered in this context. Some more complicated contexts can also be considered as context seems to play a crucial impact on the differences. Different algorithms can be experimented with for these comparisons, like Dijkstra, A*, cellular automata, etc. Besides, various contexts, like outdoor urban or indoor contexts like hospitals, shopping malls, and museums, can be experimented using similar methodologies. Notably, a self-organizing map or travelling salesman problem algorithm can be helpful for exhibition context instead of using multiple attractors with the isovist simulation. Nevertheless, this research considers pedestrians with and without the knowledge of the context and partially their destination positions, affecting the results of their differences. Hence, these findings are expected to contribute to related works for the emergency evacuation of indoor contexts, connected navigation, and wayfinding studies.

Conclusion

In conclusion, to answer the research question regarding factors influencing the qualitative and quantitative methods in this selected context, it can be arguably claimed that the goal or destination location is one of the significant factors influencing the difference between these two methodologies. This statement tends to support the second hypothesis predicted earlier in the research. Besides, the theory of the context's size, shape, and complexity with various types of obstacles also plays an essential role in creating such differences. In the quantitative analysis, the lines of sight also impact the result and, eventually, the comparison. Additionally, the time considered for qualitative and quantitative analysis might play an important role, which this research did not believe or hypothesise. The algorithm used for the quantitative method, isovist, mimics the pedestrian pattern successfully. However, only one-fourth or rarely half of the simulation often comes closer to the actual observation or the paths the pedestrian takes in real life. Hence, more information for the algorithms should be provided, like multiple attraction points or algorithms like self-organizing maps can address such issues as quantitative methods. Regardless, this research considers the pedestrian with and without knowledge of the context, which can be used in an emergency evacuation. Additionally, the use of destination or goal can be replicated as the evacuation destination, which also tends to affect the differences between these two methodologies.

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